Supplementary Information for

A self-evolving materials database that unifies 54 public sources into one searchable space of 13.5 million crystals

Zhilong Song[1,2] and Lixue Cheng[1,2,3*]

[1]Department of Chemistry, Hong Kong University of Science and Technology, Kowloon, Hong Kong 999077, China.
[2]IAS Center for AI for Scientific Discoveries, Hong Kong University of Science and Technology, Kowloon, Hong Kong 999077, China.
[3]Department of Chemical and Biological Engineering, Hong Kong University of Science and Technology, Kowloon, Hong Kong 999077, China.

*Corresponding author. E-mail: lixuecheng@ust.hk

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S1 The integration pipeline

S1.1 Adapters

Each of the 54 source databases is reached through one adapter module implementing a common protocol. An adapter yields a stream of records, each carrying the source's native identifier, a list of property values, an optional structure, a list of cross-database identifiers the source is authoritative for, and a flag stating whether the record is experimental. A property value is a tuple of canonical property name, value, unit and method string.

Two adapter flags modify the pipeline. `properties_only` forbids the source from contributing structures to the pool, which is how licence restrictions are enforced in code rather than by convention. `always_hash` forces fingerprint matching for sources whose published cross-references are not authoritative, that is, whose cited identifier names a related entry rather than the same entry.

Each source passes through three resumable stages, leaving its artefacts on disk so that a stage can be re-run without repeating the ones before it: adapt, which writes the staged property rows and the structures needing fingerprints; hash, which computes fingerprints in parallel; and match, which resolves records against the match map and separates hits from records to be admitted.

S1.2 Structural fingerprints

Structures are fingerprinted with the BAWL hasher [**?**] under the protocol used for the original deduplicated merge, so that fingerprints computed here are comparable with those already in the pool: parse the structure; reject cells with more than 500 sites; reject disordered occupancies; reduce to the primitive cell with pymatgen [**?**]; reject primitive cells with more than 300 sites; then hash. The resulting string has the form `<md5>_<space group>_<formula>`, and the space-group token is reused to complete space-group coverage for the whole pool without a second symmetry analysis. Failures are recorded per record with a reason and never abort a run.

S1.3 Two-path matching and admission

Matching proceeds in two ordered paths against a match map holding fingerprint-to-material and (source, native identifier)-to-material entries. A record is first looked up by every authoritative identifier it carries. Records still unmatched are looked up by fingerprint. A record matching nothing is admitted as a new pool material: its structure is appended to the structure blob, its metadata rows are written, and its fingerprint and identifiers are inserted into the match map, which is what

makes the enlarged substrate available to subsequent sources. The chemical formula is never used as a match criterion.

Every match writes a row recording the material, the source, the native identifier and the route taken, with `match_via` equal to `id:<database>`, `hash` or `origin`. The `origin` route records that a source contains the material as its own entry. For a source that admitted the structure this means it contributed it; for a properties-only source it means only that the source holds a corresponding record. The distinction is load-bearing for the licence position in Note S5.

S1.4 Database schema

Five tables extend the structure pool. `source_defs` holds one row per database with its licence and canonical URL. `property_defs` holds the canonical property registry. `material_properties` holds one row per value with its material, property, source, native identifier, numeric or text value and method. `material_source_refs` holds the provenance references described above. `material_curves` holds one row per stored spectrum with its type, format, source, method and an offset into a separate binary blob.

No value is ever overwritten or averaged. Several sources reporting the same property for the same material coexist as separate rows, and the interface exposes them as such. The scalar filter columns on the structure table, which exist so that common queries need not join the property table, are backfilled from `material_properties` following a fixed source priority; existing values are never overwritten, and the backfilled column is a convenience view over a record that remains complete.

S2 The domain where structural fingerprints fail

A bonding-graph fingerprint identifies a crystal by its connectivity and its primitive composition. For crystals that differ in either, this is exactly the right invariant. Enumerated framework libraries violate the premise.

Hypothetical zeolites and hypothetical metal–organic frameworks are generated by decorating a small number of parent topologies. Sibling entries therefore share a bonding graph and a primitive composition, which is the entire content of the fingerprint. Measured on the Deem zeolite set, the fingerprint path collapses 126,578 distinct frameworks onto 281 pool materials, one of which absorbs 27,007 of them. On the hypothetical metal–organic framework set, 46,891 of 135,818 hashable records collided with an earlier record and were dropped by the standard pipeline, discarding 281,346 property values.

These are not duplicates, and treating them as such is worse than losing them. Two hypothetical frameworks differing only in a functional group have their own pore geometry and their own grand-canonical carbon dioxide isotherm; routing both through one pool material would place several mutually contradictory porosities on one structure, with nothing downstream able to detect it.

We treat the failure as a property of the source rather than of the individual record. Sources whose fingerprints provably cannot separate their own entries are declared fingerprint-unreliable, their fingerprint matches are rejected at load time, and their records are admitted under a composite key pairing the fingerprint with the cell volume per site rounded to two decimals. Volume per site is invariant to the choice of unit cell and separates frameworks that a graph hash cannot. Records that still collide on the composite key are dropped rather than merged. Critically, the colliding fingerprints are withheld from the match map, so that no later source can resolve onto an ambiguous structure. Composite admission recovered 38,571 framework structures and 358,924 property values that the fingerprint path had discarded, and purged 28,914 ambiguous fingerprints that an earlier load had already registered.

The remedy is specific, not general. Any future source built by decorating a fixed topology will need the same detection and the same treatment.

S3 Provenance and the canonical property registry

Property names are mapped onto a canonical registry, each entry with a fixed unit and value kind. Mapping is not automatic: a source field enters the registry under an existing name only when the quantity is the same physical quantity, and otherwise gains its own name.

Two decisions show where the line falls. The residual of a geometric restraint optimisation reported by a structure-prediction code is registered as `structure_prediction_r_factor` rather than `r_factor`, because the crystallographic refinement residual it would otherwise join has a different meaning and an overlapping numeric range, so no downstream filter could separate the two. Framework energies quoted per tetrahedral site in $\mathrm{kJ\,mol^{-1}}$ are registered as `energy_per_t_atom` rather than folded into `total_energy_per_atom` in $\mathrm{eV\,atom^{-1}}$.

Method strings are required to be distinguishing. Where one source reports the same property for the same material under several settings, each value carries a method string that separates them, and the loader treats (native identifier, property, method) as the uniqueness key. How far this has to go is set by the source rather than by the property. Band gap carries 12,275 distinct method strings across the database, but 12,226 of them belong to the JARVIS tight-binding dataset and

take the form `QETB (tight-binding) @ 13.22 Å³/atom`: its fitting set contains many deliberately non-equilibrium geometries of the same compound, so the cell volume per atom is what separates one value from another and the method string has to carry it. The remaining 49 strings are the functional and setting labels of every other source combined. Without the volume in the string those 12,226 values would collapse onto one per material, and the fidelity gap between tight-binding and density functional theory visible in Fig. 2f of the main text would be unrecoverable.

S4 Completeness audit

The audit is deliberately independent of the loader. For each source it requires three separate equalities to hold, each able to fail on its own.

Load integrity. The row count the loader reported must equal the row count the database holds for that source. A mismatch means a load was interrupted, ran twice, or was cleared and never re-inserted. A summary table alone would never show this, because it reports only what the database currently contains.

Staging yield. Every property row the adapter staged must be either present in the database or accounted for as a native identifier that no pool material corresponds to. Staged minus loaded minus unmatched must be zero; anything else is silently dropped data.

Field coverage. Every distinct property name present in the staged data must exist in the database for that source. This is the check that catches the failure mode that matters most: a field the adapter maps but the loader never lands, which leaves row totals plausible while an entire quantity goes missing.

Sources staged on disk but absent from the source table are reported as never loaded, and sources in the registry with no staging directory are reported as not staged, so nothing can escape scrutiny by not being mentioned. Applied to the completed database, the audit reports 54 sources and no discrepancies.

Each check is computed with a single grouped scan over the property table. This is not a stylistic preference. The natural formulation, a correlated subquery counting rows per source, rescans 121 million rows once per source, takes about an hour, and evaluates each source against a different snapshot of a database other processes may be writing. Every aggregate statistic reported in this work uses grouped scans for the same reason.

S5 Licence handling and the ICSD position

Licences are enforced at three points. Sources that may not redistribute structures are marked properties-only in code, so no structure of theirs can enter the pool even if a future adapter change would otherwise admit one. Sources whose terms prohibit redistribution are excluded from any public snapshot. And the distinction between an `origin` reference as contribution and an `origin` reference as correspondence is maintained, because it is the difference between holding a licensed structure and holding a pointer to one.

The Inorganic Crystal Structure Database is the case where this matters most. Its adapter is properties-only. It annotates pool materials with chemical and mineral names and calculated densities, and its 73,823 origin references are provenance pointers marking pool materials that correspond to an ICSD collection code. Its count of admitted structures is zero, its staged structure file is empty, and no pool material lists it among its contributing sources. No FIZ Karlsruhe structure is stored or redistributed by this work.

Four sources restrict redistribution in total, contributing 31,642,922 values, or 26.0% of the database. They are searchable in a database built locally from the published pipeline and are filtered out before public release. Any statistic quoted from this work therefore has to state whether it describes the locally built database or the public snapshot; every number in the main text describes the former.

S6 Ingestion order and re-entrancy

Because each admitted structure enlarges the match map, a source ingested early is matched against a smaller substrate than the same source ingested late. Ingestion order is therefore not neutral, and Table S1 should be read with the load order of Fig. 2c in mind.

The pipeline is built to be re-entrant so that this is recoverable: re-running a source after the pool has grown is additive rather than destructive, because the loader keys property rows on (native identifier, property, method) and reference rows on (material, source, native identifier), and admitted fingerprints are already registered. Two consequences follow for anyone reproducing this work. Re-running an early source will attach references that the first pass could not resolve, and the absolute figures will therefore rise rather than change character. And a database that appears to contribute nothing new may simply have been ingested before the structures it describes had arrived.

S7 Cross-database agreement protocol

Agreement was computed for eight properties whose values are directly comparable between databases once the method is held fixed. Seven have at least one structure described by two databases and are reported; volume per atom is supplied by a single source and drops out. Extensive and convention-dependent quantities were excluded by construction, in particular total energies and prototype labels: two databases disagreeing on a total energy reflects a different reference state rather than a different answer.

For each property, all values were read in one sequential pass and grouped by material. Within a material each source was reduced to the median of its own values, so that a source reporting a quantity under several methods is not counted more than once, and the spread was taken as the range across sources, absolute for energies, magnetisation and moduli and relative for density. The tolerance within which two values are called concordant is 50 meV for band gap, 50 meV atom^{-1} for formation energy and hull distance, 0.05 μ_{B} for magnetisation, 5 GPa for the moduli and 2% for density. Source pairs sharing fewer than 200 materials are not reported. The complete pair table is Table S4.

Two cautions apply to its interpretation. Agreement between two databases is not evidence of independence: pairs whose median absolute difference is exactly zero are carrying the same underlying calculation, and the table separates them from genuinely independent pairs by that signature rather than by provenance metadata, which the sources do not always supply. And a tolerance is a convention; the distributions in Fig. 2e are given in full so that a reader preferring a different threshold can read off the corresponding figure.

S8 Cost-tier assignment

The 69 general-purpose properties of Fig. 3b were assigned by hand to four tiers according to the calculation they require: readable from a single self-consistent run; requiring a small number of extra self-consistent runs such as a slab or a strained cell; requiring linear response or finite differences over many displacements; or requiring anharmonic, transport or beyond-semilocal treatment.

Properties whose domain is one material family, such as gas uptake or mineral name, were excluded, because their coverage measures the size of that family rather than the price of the calculation. The assignment is a judgement and is listed in full in Table S3 so that it can be disputed item by item. The conclusion drawn from it in the main text, that no property requiring a

calculation of its own reaches a tenth of the pool, does not depend on the tier boundaries: it depends only on the separation between properties that fall out of a calculation already being performed and properties that do not.

S9 Census and construction of the training sets

S9.1 What counts as one task

A task is one property at one level of theory, not one property. Method strings are collapsed onto a level of theory by a fixed mapping, so that semilocal, PBEsol, SCAN, r^{2}SCAN, hybrid, meta-GGA, tight-binding and experimental values filed under a single property name become separate targets. Under that definition the 151 populated properties yield 497 candidate tasks. Counting the tasks with at least 10^6, 10^5, 10^4, 2×10^3 and 3×10^2 trainable crystals gives 13, 97, 161, 210 and 249 regression tasks respectively, over 13, 47, 77, 97 and 110 distinct properties, and 1, 15, 31, 37 and 40 classification tasks. Every structure in the pool has a stored geometry, so no task has labels without inputs; candidate structures average 15.1 sites and are capped at 500.

S9.2 Four constraints the corpus imposes on the labels

Mixed levels of theory. Of the 7.82 million crystals carrying a band gap, 81% are semilocal, 8% tight-binding and 4% each PBEsol and SCAN, with HSE06, meta-GGA, GLLB-SC and measured values making up 13 levels in total. Offsets between levels are of order 1 eV and cannot be inferred from structure, so a model trained on the pooled column learns which database supplied the row. The default policy keeps the largest level per property; a split policy trains one model per level.

Condition-dependent quantities. Some properties carry several values per crystal that are all correct. Carbon dioxide uptake is reported at four grand-canonical conditions, hydrogen uptake at five temperature and pressure combinations, decomposition energy against four different reference hulls, migration barrier by method, dimensionality and working ion, and topological class with and without spin–orbit coupling. Enumerated zeolite libraries store directional components of the Poisson ratio and stress-axis components of the Young's modulus rather than an isotropic average. Taking a median over these yields a number that is not any physical quantity, so each condition becomes its own task.

Cross-database disagreement. Across the 162 constructed sets, 5,831,192 crystals carry labels from two or more databases and 519,599 of them disagree by more than a quarter of the target's

standard deviation. The resolved label is the median, but the spread and the number of contributing sources are written alongside it, so that the disagreement can weight the loss rather than disappear into it.

Split leakage. The same compound recurs under different accession numbers, so a random split places identical chemistry on both sides and inflates the reported score in exactly the regime a completion is wanted for. Holding out whole chemical systems is the default, but it degenerates where a task occupies a single system: 135,887 hypothetical zeolites are all SiO_2 and the carbon allotrope library is all carbon, so a chemistry-level holdout leaves both the validation and the test fold empty. This happened for 23 of the 162 sets before the rule was made to fall back through formula to random and to record the level actually used. Splits are cut by material count rather than by group, because hashing groups directly sends 29% of the zeolites into a fold targeted at 10%.

S9.3 Applicability domains

Each task carries two descriptions of the structures it may be asked about. The *declared* domain is written by hand and states what the property requires of a structure: 78 tasks are pool-wide, 25 require accessible porosity, 22 a mobile working ion, 15 a gap, 4 a framework topology, 2 a magnetically ordered ground state, 2 a van der Waals gap and 2 are surface quantities held against bulk cells. Five are labelled synthesized-only, where a missing label means nobody measured it rather than that the quantity is undefined. Seven are labelled biased-labels: geometric pore descriptors are computable for any periodic structure and give a well-defined answer, usually near zero, for a dense one, so their concentration in porous materials is a sampling bias rather than a domain restriction, and the two are handled differently.

The *empirical* domain is measured from the training set: the family distribution, the number of chemical systems, the 99th percentiles of density and site count, and the elements present in every training structure. The last is the hardest constraint. Carbon dioxide uptake has never been trained on a carbon-free material; lithium migration barriers have never been trained on a lithium-free one. Disagreement between the declared and the empirical domain is itself informative: a property declared pool-wide whose labels sit in one family is biased rather than restricted.

Enforcing the domains, 35 tasks may be asked about at least 90% of the pool, 22 about 50 to 90%, 36 about 10 to 50%, 39 about 1 to 10% and 30 about less than 1%. The total number of admissible predictions is 871,097,089 against 2,194,010,874 for 162 tasks over the whole pool. One case is worth recording: a force-field bulk modulus trained on 135,817 of the pool's 136,305 zeolites

has 474 structures left to predict inside its validated domain, and its 13.4 million apparent gaps are all outside it.

The gate is structural rather than geometric. Porosity is enforced through family labels rather than through a computed pore-limiting diameter compared against the kinetic diameter of the adsorbate, which is the correct test and requires a geometry analysis over the whole pool that has not been run. A permissive mode that admits unlabelled structures whose density falls in the training band is available for candidate generation only, and has a known failure direction, since lithium metal passes a porosity density gate without having pores. Seventeen tasks are gated on a gap or a magnetic moment that the database does not hold for 75,674,942 structure-task pairs; those decisions are deferred until the corresponding tier-1 models have run.

Table S2: **The 151 properties carrying at least one value.** Ranked by the number of structures they cover. *Sources* is the number of databases supplying the property; *shared* is the number of structures for which two or more databases supply it, and is the subset on which cross-database comparison is possible.

Property	Unit	Structures	Sources	Shared
`total_energy_per_atom`	eV/atom	11,576,328	14	893,241
`formation_energy_per_atom`	eV/atom	9,872,344	14	870,683
`total_magnetization`	μB	8,071,469	8	145,353
`spacegroup_number`	–	8,018,193	21	551,369
`band_gap`	eV	7,817,982	23	452,584
`energy_above_hull`	eV/atom	7,309,591	11	376,813
`dos_at_fermi`	states/eV	5,934,625	2	24,863
`band_gap_direct`	eV	5,894,763	3	1,578
`phase_separation_energy`	eV/atom	5,894,220	1	0
`structure_prototype`	–	2,794,238	11	29,879
`density`	g/cm^3	2,494,079	10	42,421
`volume_per_atom`	Å^3/atom	2,275,947	1	0
`max_residual_force`	eV/Å	2,005,836	1	0
`max_residual_stress`	eV/Å^3	2,005,836	1	0
`magnetization_per_atom`	μB/atom	938,459	1	0
`dimensionality`	–	895,781	12	14,052
`decomposition_energy`	eV/atom	617,476	3	839
`largest_cavity_diameter`	Å	473,815	7	16,043
`pore_limiting_diameter`	Å	473,282	6	16,043
`void_fraction`	–	455,303	6	9,747
`surface_area_gravimetric`	m^2/g	430,332	5	6,193
`co2_uptake`	mol/kg	415,108	2	0
`pore_volume`	cm^3/g	299,394	2	0
`n2_uptake`	mol/kg	295,508	2	0

Table S2, continued

Property	Unit	Structures	Sources	Shared
co2_n2_selectivity	–	287,610	1	0
working_capacity_vacuum_swing	mol/kg	287,610	1	0
working_capacity_temperature_swing	mol/kg	287,610	1	0
co2_heat_of_adsorption	kJ/mol	286,229	1	0
n2_heat_of_adsorption	kJ/mol	285,827	1	0
is_gap_direct	–	243,327	3	16,244
measurement_temperature	K	238,161	1	0
experimentally_observed	–	234,892	2	100
dielectric_constant_electronic	–	226,290	5	11,359
fermi_energy	eV	215,053	2	186
is_stable	–	211,854	1	0
magnetization_per_volume	μB/Å^3	211,854	1	0
is_metal	–	211,753	2	95
bulk_modulus_vrh	GPa	196,615	6	15,275
poisson_ratio	–	195,059	5	15,252
shear_modulus_vrh	GPa	192,267	5	15,272
magnetic_ordering	–	182,470	4	11,788
surface_area_volumetric	m^2/cm^3	167,537	5	8,355
chemical_name	–	165,051	3	4,638
oxidation_states	–	158,465	1	0
average_oxidation_states	–	158,465	1	0
bulk_modulus_voigt	GPa	156,720	2	0
bulk_modulus_reuss	GPa	156,660	2	0
sound_velocity_longitudinal	m/s	156,645	2	0
shear_modulus_voigt	GPa	155,061	2	0
dielectric_constant_total	–	154,893	5	4,265
sound_velocity_transverse	m/s	153,763	2	0
decomposes_to	–	153,455	1	0
shear_modulus_reuss	GPa	151,750	2	0

Table S2, continued

Property	Unit	Structures	Sources	Shared
framework_density	T/1000 Å³	148,195	3	0
youngs_modulus	GPa	136,108	2	0
energy_per_t_atom	kJ/mol	136,008	1	0
cbm	eV	123,288	1	0
vbm	eV	123,288	1	0
predicted_synthesizable	–	94,797	1	0
magnetic_species	–	80,996	1	0
topology_class	–	80,342	2	47,043
topology_invariant	–	70,967	2	23,299
phonon_min_frequency	THz	44,432	4	12,416
has_imaginary_phonon	–	44,432	4	12,416
phonon_max_frequency	THz	44,408	4	12,416
symmetry_indicators	–	41,916	1	0
seebeck_coefficient	μV/K	40,378	2	294
power_factor	μW/cm·K²	40,375	2	293
largest_free_path_diameter	Å	36,602	2	2,271
pore_channel_count	–	27,089	1	0
lattice_thermal_conductivity	W/m·K	26,507	2	1,937
debye_temperature	K	25,997	2	5,333
heat_of_formation	eV/atom	24,570	1	0
elastic_anisotropy	–	24,430	1	0
heat_capacity	J/mol·K	23,135	2	973
framework_bond_dissociation_energy	eV/atom	20,194	1	0
nmr_efg_max	V/m²	19,740	1	0
dielectric_constant_ionic	–	19,484	3	4,167
average_voltage	V	19,280	1	0
capacity_gravimetric	mAh/g	19,280	1	0
capacity_volumetric	mAh/cm³	19,280	1	0
energy_density_gravimetric	Wh/kg	19,280	1	0

Table S2, continued

Property	Unit	Structures	Sources	Shared
energy_density_volumetric	Wh/L	19,280	1	0
max_voltage_step	V	19,280	1	0
num_voltage_steps	–	19,280	1	0
electrode_stability_charged	eV/atom	19,280	1	0
electrode_stability_discharged	eV/atom	19,280	1	0
electrode_volume_change	fraction	19,280	1	0
working_ion	–	19,280	1	0
battery_formula	–	19,280	1	0
spillage	–	19,113	2	88
refractive_index	–	19,080	3	731
slme	%	16,815	2	64
structure_prediction_r_factor	–	12,415	1	0
piezoelectric_modulus	C/m²	12,222	2	2,441
thermal_conductivity	W/m·K	10,565	1	0
thermal_expansion	1/K	10,563	1	0
mineral_name	–	9,632	6	4,757
gllbsc_derivative_discontinuity	eV	8,236	1	0
eos_bulk_modulus	GPa	8,060	1	0
ar_uptake	mol/kg	7,898	1	0
lattice_thermal_conductivity_coherence	W/m·K	7,689	1	0
lattice_thermal_conductivity_particle	W/m·K	7,678	1	0
born_effective_charge_max	e	7,558	1	0
effective_mass_electron	m_e	7,324	3	356
effective_mass_hole	m_e	7,309	3	360
exfoliation_energy	meV/atom	6,395	3	636
coordination_environments	–	6,108	1	0
continuous_symmetry_measure	–	6,108	1	0
piezoelectric_max_direction	–	5,926	1	0
magnetic_polarization	T	5,309	1	0

Table S2, continued

Property	Unit	Structures	Sources	Shared
tc_superconducting	K	5,091	4	2,431
birefringence	–	4,796	1	0
interlayer_binding_energy	meV/Å²	4,126	2	0
work_function	eV	3,730	2	0
migration_barrier	eV	3,598	1	0
pearson_symbol	–	3,312	2	8
h2_uptake	g/L	2,767	1	0
superconductor_class	–	2,621	1	0
shg_coefficient	pm/V	2,566	1	0
xe_uptake	mol/kg	2,549	1	0
kr_uptake	mol/kg	2,549	1	0
magnetic_anisotropy_constant	MJ/m³	1,489	1	0
easy_axis	–	1,489	1	0
relative_energy_to_diamond	eV/atom	1,452	1	0
electron_phonon_coupling	–	1,431	1	0
phonon_log_average_frequency	K	1,431	1	0
hardness_vickers	GPa	1,414	1	0
measurement_pressure	GPa	1,303	2	32
mohs_hardness	–	1,183	1	0
dynamically_stable	–	1,023	1	0
layer_group_number	–	1,023	1	0
bulk_modulus_eos	GPa	502	1	0
bulk_modulus_pressure_derivative	–	502	1	0
d_band_center	eV	485	1	0
strukturbericht	–	473	1	0
surface_energy	J/m²	237	1	0
has_reconstructed_surface	–	237	1	0
surface_anisotropy	–	230	1	0
wulff_shape_factor	–	230	1	0

Table S2, continued

Property	Unit	Structures	Sources	Shared
r_factor	–	183	1	0
measured_chemistry	–	176	1	0
elf_max	–	145	1	0
grain_boundary_energy	J/m²	106	1	0
work_of_separation	J/m²	106	1	0
grain_boundary_types	–	106	1	0
altermagnetic_splitting_max	eV	103	1	0
altermagnetic_splitting_avg	eV	103	1	0
altermagnetic_split_fraction	%	103	1	0
ionic_conductivity	S/cm	49	1	0
curie_temperature	K	6	1	0

Table S4: **Pairwise agreement between databases.** Every pair of databases sharing at least 200 structures for a comparable property. Δ is the difference between the two databases in the order listed. A median signed difference close to the median absolute difference indicates that nearly all of the disagreement carries one sign.

Property	Pair	Shared	med. Δ
band_gap	matbench-discovery / alexandria	138,961	+0
band_gap	oqmd / alexandria	113,998	+0
band_gap	materials-project / alexandria	94,900	+0
band_gap	materials-project / jarvis-dft	72,603	+0
band_gap	oqmd / aflow	59,954	+0
band_gap	jarvis-dft / alexandria	58,746	+0
band_gap	materiae / materials-project	55,909	+0
band_gap	materials-project / aflow	54,542	+0

Table S4, continued

Property	Pair	Shared	med. Δ
band_gap	materials-project / oqmd	54,484	+0
band_gap	matbench-discovery / oqmd	51,166	+0
band_gap	alexandria / aflow	39,823	+0
band_gap	jarvis-dft / aflow	37,819	+0
band_gap	jarvis-dft / oqmd	37,647	+0
band_gap	materiae / aflow	36,933	+0
band_gap	materiae / jarvis-dft	35,632	+0
band_gap	alexandria / jarvis-qetb	28,563	+0
band_gap	materiae / oqmd	25,921	+0
band_gap	materiae / alexandria	24,870	+0
band_gap	oqmd / jarvis-qetb	23,084	+0
band_gap	jarvis-dft / jarvis-qetb	20,578	+0
band_gap	snumat / materials-project	16,149	-0.777
band_gap	snumat / aflow	13,951	-0.671
band_gap	materials-project / cmr-static	13,415	+0.176
band_gap	cmr-static / aflow	12,323	+0
band_gap	gnome / alexandria	11,218	-0.004
band_gap	materiae / cmr-static	11,164	+0.429
band_gap	cmr-static / jarvis-dft	10,992	+0
band_gap	snumat / materiae	10,807	-0.784
band_gap	snumat / jarvis-dft	10,541	-0.708
band_gap	cmr-static / oqmd	10,116	+0
band_gap	snumat / oqmd	9,359	-0.672
band_gap	materials-project / jarvis-qetb	8,763	+0
band_gap	matbench-discovery / jarvis-dft	8,668	+0
band_gap	snumat / alexandria	7,574	-0.773
band_gap	cmr-static / alexandria	7,381	+0
band_gap	matbench-discovery / aflow	7,271	+0
band_gap	aflow / jarvis-qetb	7,250	+0

Table S4, continued

Property	Pair	Shared	med. Δ
band_gap	matbench-discovery / materials-project	7,162	+0
band_gap	matbench-discovery / jarvis-qetb	6,065	+0
band_gap	snumat / cmr-static	5,428	+0.120
band_gap	materiae / jarvis-qetb	4,731	+0
band_gap	cmr-static / jarvis-qetb	3,051	+0
band_gap	materials-cloud-archive-static / alexandria	2,972	+0
band_gap	gnome / oqmd	2,866	-0.004
band_gap	matbench-discovery / materiae	2,327	+0
band_gap	materials-cloud-archive-static / materials-project	2,050	+0
band_gap	2dmatpedia / alexandria	1,543	+0
band_gap	materials-cloud-archive-static / jarvis-dft	1,170	+0.0444
band_gap	c2db / alexandria	1,125	+0
band_gap	snumat / jarvis-qetb	1,027	-1.11
band_gap	matbench-discovery / cmr-static	975	+0
band_gap	2dmatpedia / materials-project	953	-0.021
band_gap	2dmatpedia / jarvis-dft	879	+0
band_gap	c2db / bidb	848	+0.00752
band_gap	materiae / materials-cloud-archive-static	825	+0.143
band_gap	matbench-discovery / materials-cloud-archive-static	761	-0.005
band_gap	2dmatpedia / mc2d	732	+0
band_gap	c2db / jarvis-qetb	721	+0
band_gap	mc2d / alexandria	707	+0
band_gap	jarvis-specialized-structures / 2dmatpedia	671	+0
band_gap	2dmatpedia / jarvis-qetb	658	+0
band_gap	materials-cloud-archive-static / oqmd	631	+0.150
band_gap	bidb / alexandria	623	+0
band_gap	matbench-discovery / gnome	621	+0.007
band_gap	materials-cloud-archive-static / aflow	614	-0.0612
band_gap	jarvis-specialized-structures / jarvis-dft	548	+0

Table S4, continued

Property	Pair	Shared	med. Δ
band_gap	c2db / 2dmatpedia	545	+0
band_gap	jarvis-specialized-structures / alexandria	533	+0
band_gap	matbench-discovery / snumat	525	+0.773
band_gap	2dmatpedia / aflow	511	-0.001
band_gap	materiae / 2dmatpedia	495	+0.159
band_gap	mc2d / materials-project	477	-0.0171
band_gap	mc2d / jarvis-dft	431	+0
band_gap	2dmatpedia / bidb	414	+0
band_gap	jarvis-specialized-structures / materials-project	386	-0.0
band_gap	2dmatpedia / oqmd	381	-0.0
band_gap	jarvis-specialized-structures / mc2d	381	-0.0133
band_gap	c2db / oqmd	376	+0
band_gap	mc2d / aflow	339	+0
band_gap	snumat / materials-cloud-archive-static	326	-0.529
band_gap	materiae / mc2d	293	+0.220
band_gap	matbench-discovery / c2db	287	+0
band_gap	materiae / jarvis-specialized-structures	278	+0.165
band_gap	snumat / 2dmatpedia	273	-0.6
band_gap	jarvis-specialized-structures / aflow	271	+0
band_gap	jarvis-specialized-structures / jarvis-qetb	260	-0.0
band_gap	matbench-discovery / 2dmatpedia	258	+0.0221
band_gap	mc2d / bidb	240	-2.869e-0
band_gap	c2db / jarvis-dft	239	+0
band_gap	bidb / jarvis-qetb	238	-0.112
band_gap	jarvis-specialized-structures / c2db	233	+0
band_gap	c2db / mc2d	230	+0.0222
band_gap	jarvis-specialized-structures / bidb	227	+0
band_gap	snumat / mc2d	223	-0.730
band_gap	bidb / oqmd	203	+0

Table S4, continued

Property	Pair	Shared	med. Δ
bulk_modulus_vrh	materials-project / jarvis-dft	12,096	+4.65
bulk_modulus_vrh	materials-project / aflow	5,617	-0.21
bulk_modulus_vrh	jarvis-dft / aflow	4,949	-4.68
density	aflow / icsd	37,882	+0.0422
density	amcsd / aflow	2,828	-0.0410
density	core-mof / qmof	2,028	+0.0477
density	amcsd / icsd	1,533	-0.000220
density	icsd / webmineral	1,219	-0.00571
density	aflow / webmineral	613	+0.0335
energy_above_hull	matbench-discovery / alexandria	138,961	+0.0113
energy_above_hull	oqmd / alexandria	114,078	+
energy_above_hull	materials-project / alexandria	94,900	+
energy_above_hull	materials-project / jarvis-dft	72,603	+1.64
energy_above_hull	jarvis-dft / alexandria	58,746	-1.61
energy_above_hull	materials-project / oqmd	54,617	+
energy_above_hull	matbench-discovery / oqmd	51,180	+0.0128
energy_above_hull	jarvis-dft / oqmd	37,765	-1.3
energy_above_hull	odbx / alexandria	24,507	+
energy_above_hull	odbx / materials-project	22,580	+
energy_above_hull	odbx / jarvis-dft	19,662	+1.49
energy_above_hull	odbx / oqmd	11,924	+2.888e-0
energy_above_hull	matbench-discovery / jarvis-dft	8,668	+1.27
energy_above_hull	matbench-discovery / materials-project	7,162	+0.00923
energy_above_hull	matbench-discovery / odbx	3,472	+0.0242
energy_above_hull	odbx / cmr-static	3,392	-0.054
energy_above_hull	cmr-static / alexandria	2,396	+0.0703
energy_above_hull	cmr-static / jarvis-dft	2,389	+1.28
energy_above_hull	materials-project / cmr-static	2,332	-0.0693
energy_above_hull	novomag / materials-project	1,853	+

Table S4, continued

Property	Pair	Shared	med. Δ
energy_above_hull	cmr-static / oqmd	1,716	+0.0604
energy_above_hull	novomag / jarvis-dft	1,441	+2.48
energy_above_hull	novomag / alexandria	1,414	+
energy_above_hull	novomag / oqmd	884	+0.000283
energy_above_hull	novomag / odbx	653	+
energy_above_hull	bidb / alexandria	626	+0.00416
energy_above_hull	matbench-discovery / cmr-static	540	-0.0323
energy_above_hull	jarvis-specialized-structures / alexandria	527	+0.00711
energy_above_hull	jarvis-specialized-structures / jarvis-dft	412	+1.43
energy_above_hull	jarvis-specialized-structures / materials-project	383	+0.0144
energy_above_hull	jarvis-specialized-structures / odbx	317	+0.000736
energy_above_hull	matbench-discovery / novomag	258	+0.00910
energy_above_hull	novomag / cmr-static	239	-0.0386
energy_above_hull	jarvis-specialized-structures / bidb	229	+0.00124
energy_above_hull	odbx / bidb	215	-0.00275
energy_above_hull	bidb / oqmd	204	+0.00290
formation_energy_per_atom	alexandria / aflow	436,461	-0.00507
formation_energy_per_atom	oqmd / aflow	148,626	-0.00036
formation_energy_per_atom	matbench-discovery / alexandria	138,961	+0.00135
formation_energy_per_atom	oqmd / alexandria	114,078	+0.00118
formation_energy_per_atom	materials-project / alexandria	94,900	+4.145e-0
formation_energy_per_atom	materials-project / jarvis-dft	72,603	+0.0706
formation_energy_per_atom	jarvis-dft / alexandria	58,746	-0.010
formation_energy_per_atom	materials-project / oqmd	54,617	+0.0506
formation_energy_per_atom	matbench-discovery / oqmd	51,180	+0.00308
formation_energy_per_atom	aflow / jarvis-qetb	46,079	+0.149
formation_energy_per_atom	jarvis-dft / oqmd	37,765	-0.00070
formation_energy_per_atom	jarvis-dft / aflow	31,511	+0.0328
formation_energy_per_atom	alexandria / jarvis-qetb	28,563	+0.15

Table S4, continued

Property	Pair	Shared	med. Δ
formation_energy_per_atom	materials-project / aflow	27,180	+0.17
formation_energy_per_atom	matbench-discovery / aflow	25,191	-0.00345
formation_energy_per_atom	odbx / alexandria	24,543	-0.002
formation_energy_per_atom	oqmd / jarvis-qetb	23,088	+0.218
formation_energy_per_atom	odbx / materials-project	22,913	-0.0647
formation_energy_per_atom	jarvis-dft / jarvis-qetb	20,578	+0.156
formation_energy_per_atom	odbx / jarvis-dft	19,846	+0.0155
formation_energy_per_atom	gnome / alexandria	14,709	+0.00968
formation_energy_per_atom	odbx / oqmd	12,235	+
formation_energy_per_atom	odbx / aflow	12,111	+0.0402
formation_energy_per_atom	odbx / cmr-static	10,253	+0.0108
formation_energy_per_atom	materials-project / jarvis-qetb	8,766	+0.154
formation_energy_per_atom	matbench-discovery / jarvis-dft	8,668	-0.0113
formation_energy_per_atom	matbench-discovery / materials-project	7,162	+0.000889
formation_energy_per_atom	cmr-static / alexandria	6,592	-0.0132
formation_energy_per_atom	materials-project / cmr-static	6,258	+0.0826
formation_energy_per_atom	cmr-static / jarvis-dft	6,092	-0.0100
formation_energy_per_atom	matbench-discovery / jarvis-qetb	6,065	+0.207
formation_energy_per_atom	odbx / jarvis-qetb	5,512	+0.144
formation_energy_per_atom	cmr-static / aflow	4,667	+0.00880
formation_energy_per_atom	gnome / oqmd	4,265	+0.0323
formation_energy_per_atom	cmr-static / oqmd	3,498	+
formation_energy_per_atom	matbench-discovery / odbx	3,479	+0.00339
formation_energy_per_atom	cmr-static / jarvis-qetb	2,457	+0.0783
formation_energy_per_atom	matbench-discovery / cmr-static	1,093	-0.0035
formation_energy_per_atom	matbench-discovery / gnome	800	-0.00657
formation_energy_per_atom	oqmd / carolina	645	-1.07
formation_energy_per_atom	jarvis-specialized-structures / alexandria	570	-0.117
formation_energy_per_atom	jarvis-specialized-structures / jarvis-dft	423	-0.0727

Table S4, continued

Property	Pair	Shared	med. Δ
formation_energy_per_atom	jarvis-specialized-structures / materials-project	398	-0.240
formation_energy_per_atom	jarvis-specialized-structures / odbx	325	-0.162
formation_energy_per_atom	jarvis-specialized-structures / jarvis-qetb	283	+0.0335
formation_energy_per_atom	jarvis-specialized-structures / aflow	232	-0.0422
shear_modulus_vrh	materials-project / jarvis-dft	12,096	+2.98
shear_modulus_vrh	materials-project / aflow	5,617	-0.16
shear_modulus_vrh	jarvis-dft / aflow	4,949	-2.76
total_magnetization	materials-project / alexandria	94,904	-3.4e-0
total_magnetization	materials-project / jarvis-dft	67,436	-3.3e-0
total_magnetization	jarvis-dft / alexandria	55,521	+1e-0
total_magnetization	elementa / alexandria	10,929	-1e-0
total_magnetization	novomag / materials-project	3,257	-0.0388
total_magnetization	novomag / alexandria	2,440	-0.00659
total_magnetization	novomag / jarvis-dft	2,138	-0.08
total_magnetization	2dmatpedia / alexandria	1,543	+
total_magnetization	2dmatpedia / materials-project	953	+
total_magnetization	2dmatpedia / jarvis-dft	766	+
total_magnetization	jarvis-specialized-structures / 2dmatpedia	671	+
total_magnetization	elementa / jarvis-dft	562	+
total_magnetization	jarvis-specialized-structures / alexandria	527	+
total_magnetization	jarvis-specialized-structures / materials-project	383	+
total_magnetization	elementa / materials-project	362	+2.5e-0
total_magnetization	jarvis-specialized-structures / jarvis-dft	361	+

Table S1 The 54 integrated source databases. Property values, distinct properties and annotated structures are counted in the completed database. *Matched* references were resolved by identifier or fingerprint; *own* references record that the database contains the structure as its own entry. A dagger marks the four databases whose terms prohibit redistribution.

Database	Licence	Values	Props	Structures	Matched	Own
General-purpose DFT databases						
alexandria	Alexandria dataset terms	60,588,138	10	5,894,222	5,200,785	1,514,488
aflow†	AFLOW academic non-commerc	18,464,119	14	2,275,947	1,401,895	1,650,288
elementa†	CC BY-NC 4.0	8,023,344	4	2,005,836	2,005,836	0
oqmd	CC BY 4.0	6,086,500	5	1,229,047	1,022,154	255,085
gnome†	CC BY-NC 4.0	4,897,901	5	554,056	6,453	547,603
materials-project	Materials Project terms	4,291,860	61	211,854	213,332	3,157
jarvis-qetb	NIST/JARVIS dataset terms	2,524,250	3	191,606	716,844	124,598
matbench-discovery	CC BY 4.0	1,734,187	5	401,437	290,249	136,041
jarvis-dft	NIST/JARVIS dataset terms	1,569,924	26	120,277	117,827	14,986
carolina	CC BY 4.0	428,272	2	210,000	170,438	43,801
odbx	ODBX provider terms and so	287,006	2	142,987	42,056	103,138
mc3d	Materials Cloud record lic	70,508	1	68,487	63,775	6,733
Experimental structure catalogues						
cod	CC0	354,350	4	265,903	41,249	238,839
icsd†	FIZ Karlsruhe ICSD licence	257,558	3	77,041	49,579	73,823
Specialised property libraries						
materiae	Materiae site terms; cite	589,993	6	67,688	63,612	4,769
cmr-static	CC BY-SA 4.0	305,346	10	192,185	174,387	33,044
snumat	Figshare record licence	170,893	4	21,115	18,187	3,273
phonix	CC BY 4.0	155,192	8	22,733	28,247	2,171
materials-cloud-archive-static	record-specific open licen	143,153	11	68,547	69,224	2,811
csllm-negatives	CC BY 4.0	98,303	1	94,797	98,303	0
phonondb	CC BY 4.0	86,616	7	15,245	13,896	1,700
novomag	CC BY 4.0	38,884	10	5,309	3,861	1,573
3dsc	3DSC repository licence an	36,531	2	3,123	28,324	0
2dmatpedia	2DMatPedia terms	30,583	5	6,491	2,246	4,275
spse	CC BY 4.0 record; underlyi	29,953	2	3,599	5,005	0
jarvis-specialized-structures	NIST/JARVIS dataset terms	26,154	22	3,443	3,562	574
amcsd	AMCSD terms	24,639	2	7,790	13,518	1,203
c2db	CC BY-SA 4.0	14,120	4	3,493	1,277	2,253
mc2d	Materials Cloud record lic	13,202	4	3,164	1,319	1,873
omdb-organic	OMDB terms	12,013	1	11,975	125	11,888
sacada	SACADA site terms and sour	10,349	7	1,532	27	1,508
bidb	2DHub dataset terms	9,405	7	1,023	1,032	17
superband	CC BY 4.0 data; MIT softwa	8,985	2	3,147	7,241	7
aflow-prototypes	Apache-2.0	6,251	5	1,788	1,799	0
webmineral	Webmineral site and cited-	4,698	4	1,320	1,330	0
openmx-2d-ab2	OpenMX database and source	4,305	2	2,151	382	1,797
hex-high-pressure	CC0	1,456	2	241	694	37
rruff	RRUFF terms and source-cit	1,302	2	829	940	123
amdb	AMDB terms and source-publ	622	6	103	104	0
hadb	HADB terms and source-publ	492	4	123	18	105
tcod	CC0	357	3	2,803	856	2,227
obelix	CC BY 4.0	112	2	60	61	0
Porous and framework materials						
mof-carbon-capture	CC BY 4.0	5,454,666	13	287,610	0	287,610
deem-zeolites	CC BY 4.0	2,977,166	15	136,008	0	136,008

Table S3 Assignment of the 69 general-purpose properties to calculation tiers. The assignment is a hand judgement about what a property costs to obtain and is listed in full so that it can be disputed item by item. Properties whose domain is a single material family are excluded.

Tier	Properties
1 single self-consistent run median coverage 4.56%	`total_energy_per_atom`, `formation_energy_per_atom`, `energy_above_hull`, `band_gap`, `total_magnetization`, `magnetization_per_atom`, `is_metal`, `is_stable`, `magnetic_ordering`, `dos_at_fermi`, `fermi_energy`, `decomposition_energy`, `heat_of_formation`, `phase_separation_energy`, `magnetization_per_volume`, `band_gap_direct`, `is_gap_direct`, `cbm`, `vbm`
2 a few extra self-consistent runs median coverage 0.029%	`work_function`, `effective_mass_electron`, `effective_mass_hole`, `surface_energy`, `exfoliation_energy`, `interlayer_binding_energy`, `d_band_center`, `spillage`, `grain_boundary_energy`, `work_of_separation`
3 linear response or finite differences median coverage 1.06%	`bulk_modulus_vrh`, `shear_modulus_vrh`, `youngs_modulus`, `poisson_ratio`, `elastic_anisotropy`, `bulk_modulus_voigt`, `bulk_modulus_reuss`, `shear_modulus_voigt`, `shear_modulus_reuss`, `sound_velocity_transverse`, `sound_velocity_longitudinal`, `debye_temperature`, `dielectric_constant_total`, `dielectric_constant_electronic`, `dielectric_constant_ionic`, `born_effective_charge_max`, `piezoelectric_modulus`, `phonon_max_frequency`, `phonon_min_frequency`, `has_imaginary_phonon`, `eos_bulk_modulus`, `bulk_modulus_pressure_derivative`
4 anharmonic, transport or beyond-semilocal median coverage 0.0588%	`lattice_thermal_conductivity`, `lattice_thermal_conductivity_particle`, `lattice_thermal_conductivity_coherence`, `thermal_conductivity`, `thermal_expansion`, `heat_capacity`, `seebeck_coefficient`, `power_factor`, `slme`, `shg_coefficient`, `birefringence`, `refractive_index`, `gllbsc_derivative_discontinuity`, `tc_superconducting`, `electron_phonon_coupling`, `phonon_log_average_frequency`, `curie_temperature`, `migration_barrier`